# Type MS · Explain eye defects and how they are corrected or treated

## Answer 1 of 11 · Question 8 part b.iii · 2 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

| Question number | | | | | |
|---|---|---|---|---|---|
| 8 | (a) | (i) | • arrow heads showing direction of light rays;<br>• bent at cornea;<br>• bent at lens;<br>• cross over in eyeball;<br>• inverted image on retina; | | 5 marks |
| | | (ii) | transmits/carries impulses from the retina/rod and cones;<br>to the brain; | | 2 marks |
| | (b) | (i) | behind the retina/fovea; | | 1 mark |
| | | (ii) | eyeball too short / eye lens not powerful enough/too flat/not convex enough; | | 1 mark |
| | | (iii) | use a convex / converging lens;<br>to converge/refract/bend rays of light; | | 2 mark |
| | (c) | | • infection;<br>• rejection; | | 2 marks |

Total 13 marks

## Type MS · Explain eye defects and how they are corrected or treated

### Answer 1 of 11 · Question 8 part b.iii · 2 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

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Type MS · Explain eye defects and how they are corrected or treated

Answer 2 of 11 · Question 3 part c · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 3 | (a) | (i) | A (cornea and lens);<br><br>B is incorrect as the iris does not play a role in focusing light<br>C is incorrect as the iris does not play a role in focusing light<br>D is incorrect as the pupil does not play a role in focusing light | | 1 |
| | (b) | (i) | Points as follows:<br>150:2.7<br>200:3.2<br>250:2.3; | | 1 |
| Clip (b)(i) with (b)(ii) and (b)(iii) | | (ii) | Line of best fit missing out point at 150:2.7 | Ignore lines that extended beyond points | 1 |
| | | (iii) | 150 lux/150:2.7/2.7 diameter; | | 1 |
| | | | does not follow the pattern of other points/trend/not on the line of best fit; | | 1 |
| | | (iv) | As light intensity increases diameter decreases; | ORA | 1 |
| | (c) | | Idea that distant/far away objects are blurred/difficult to see/ORA;<br>as (light) rays do not meet on the retina/focused in front of retina/meet before the retina/light rays refracted too much; | Reject can't see | 1<br>1 |

**Total question 3 = 8 marks**

## Type MS · Explain eye defects and how they are corrected or treated

### Answer 2 of 11 · Question 3 part c · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Explain eye defects and how they are corrected or treated

Answer 3 of 11 · Question 1 part c · 3 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| **1(a)** | line labelled A pointing to retina;<br>line labelled B pointing to ciliary muscles;<br>line labelled C pointing to iris;<br>line labelled D pointing to optic nerve;<br>line labelled E pointing to eyelid; | | 5 |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **1(b)** | bending at cornea;<br>bending at lens;<br>image formed on retina; | | **3** |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **1(c)** | long sight/myopia;<br>short sight/hypermetropia;<br>astigmatism; | | 3 |

Total 11 marks

Type MS · Explain eye defects and how they are corrected or treated

Answer 3 of 11 · Question 1 part c · 3 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

**Paper 2: Subject Human Biology**

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|

## Type MS · Explain eye defects and how they are corrected or treated

### Answer 4 of 11 · Question 8 part b.i · 1 mark

2021 NOV · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 8 | (a) | (i) | • arrow heads showing direction of light rays;<br>• bent at cornea;<br>• bent at lens;<br>• cross over in eyeball;<br>• inverted image on retina; | | 5 marks |
| | | (ii) | transmits/carries impulses from the retina/rod and cones;<br>to the brain; | | 2 marks |
| | (b) | (i) | behind the retina/fovea; | | 1 mark |
| | | (ii) | eyeball too short / eye lens not powerful enough/too flat/not convex enough; | | 1 mark |
| | | (iii) | use a convex / converging lens;<br>to converge/refract/bend rays of light; | | 2 mark |
| | (c) | | • infection;<br>• rejection; | | 2 marks |

Total 13 marks

## Type MS · Explain eye defects and how they are corrected or treated

### Answer 4 of 11 · Question 8 part b.i · 1 mark

2021 NOV · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

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## Type MS · Explain eye defects and how they are corrected or treated

### Answer 5 of 11 · Question 8 part b.ii · 1 mark

2021 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 8 | (a) | (i) | • arrow heads showing direction of light rays;<br>• bent at cornea;<br>• bent at lens;<br>• cross over in eyeball;<br>• inverted image on retina; | | 5 marks |
| | | (ii) | transmits/carries impulses from the retina/rod and cones;<br>to the brain; | | 2 marks |
| | (b) | (i) | behind the retina/fovea; | | 1 mark |
| | | (ii) | eyeball too short / eye lens not powerful enough/too flat/not convex enough; | | 1 mark |
| | | (iii) | use a convex / converging lens;<br>to converge/refract/bend rays of light; | | 2 mark |
| | (c) | | • infection;<br>• rejection; | | 2 marks |

Total 13 marks

## Type MS · Explain eye defects and how they are corrected or treated

### Answer 5 of 11 · Question 8 part b.ii · 1 mark

2021 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

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Type MS · Explain eye defects and how they are corrected or treated

Answer 6 of 11 · Question 6 part c · 2 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 6 | (a) | (i) | • A=rods/cones/retina;<br>• detects light;<br>• converts light energy into electrical energy;<br>• (changed into (nerve) impulses;<br>• B = transmits (nerve) impulses/electrical signals;<br>• from (retina) to brain; | At least 1 mark from A and B to get full marks | 4 |
| | | (ii) | • bent inwards at cornea/lens;<br>• rays pass through the cornea and converge after the lens;<br>• rays cross in eye (after lens);<br>• inverted image/arrow; | Do not accept bending outwards for MP1 and MP2<br><br>Mp3- do not accept rays crossing before the retina | 4 |
| | (b) | | • use both eyes and one eye;<br>• use of eye patch/eq;<br>• ball thrown same distance/stated distance;;<br>• ball thrown many times;<br>• count how many times ball is caught;<br>• same type/size of ball; | | 4 |
| | (c) | | • misshapen/eq. cornea/lens;<br>• light not focussed correctly/ light scatters/light not refracted properly;<br>• blurred image; | | 2 |
| | | | | | **Total =14** |

## Type MS · Explain eye defects and how they are corrected or treated

### Answer 6 of 11 · Question 6 part c · 2 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

## Type MS · Explain eye defects and how they are corrected or treated

### Answer 7 of 11 · Question 8 part c · 2 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 8 | (a) | (i) | • arrow heads showing direction of light rays;<br>• bent at cornea;<br>• bent at lens;<br>• cross over in eyeball;<br>• inverted image on retina; | | 5 marks |
| | | (ii) | transmits/carries impulses from the retina/rod and cones;<br>to the brain; | | 2 marks |
| | (b) | (i) | behind the retina/fovea; | | 1 mark |
| | | (ii) | eyeball too short / eye lens not powerful enough/too flat/not convex enough; | | 1 mark |
| | | (iii) | use a convex / converging lens;<br>to converge/refract/bend rays of light; | | 2 mark |
| | (c) | | • infection;<br>• rejection; | | 2 marks |

Total 13 marks

## Type MS · Explain eye defects and how they are corrected or treated

### Answer 7 of 11 · Question 8 part c · 2 marks

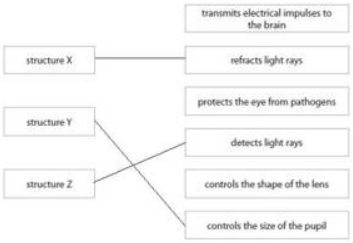
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## Type MS · Explain eye defects and how they are corrected or treated

### Answer 8 of 11 · Question 1 part b.i · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 1 | (a) | (i) |  | | 3 |
| | | (ii) | A;<br>B, C and D do not describe stereoscopic vision. | | 1 |
| | (b) | (i) | build-up of protein/cloudy/opaque;<br>in the lens; | | 2 |
| | | (ii) | surgery/operation to replace the lens/lens replacement; | | 1 |
| | | (iii) | 39 000000 x 43 ÷ 100;<br>16.77 million/16.8 million/16 770 000; | | 2 |
| | | (iv) | vitamin A; | | 1 |

Total 10 marks

## Type MS · Explain eye defects and how they are corrected or treated

### Answer 8 of 11 · Question 1 part b.i · 2 marks

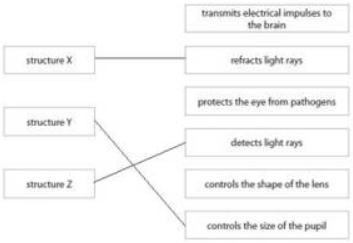
2023 JAN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Explain eye defects and how they are corrected or treated

### Answer 9 of 11 · Question 1 part b.ii · 1 mark

2023 JAN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 1 | (a) | (i) |  | | 3 |
| | | (ii) | A;<br>B, C and D do not describe stereoscopic vision. | | 1 |
| | (b) | (i) | build-up of protein/cloudy/opaque;<br>in the lens; | | 2 |
| | | (ii) | surgery/operation to replace the lens/lens replacement; | | 1 |
| | | (iii) | 39 000$^{0}$000 x 43 ÷ 100;<br>16.77 million/16.8 million/16 770 000; | | 2 |
| | | (iv) | vitamin A; | | 1 |

Total 10 marks

## Type MS · Explain eye defects and how they are corrected or treated

### Answer 9 of 11 · Question 1 part b.ii · 1 mark

2023 JAN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Explain eye defects and how they are corrected or treated

Answer 10 of 11 · Question 3 part c · 1 mark

2024 JUN · Paper 02 · find part c in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 3 (a) | Smoking cigarettes;<br>Causes (lung) cancer/emphysema/bronchitis/heart disease/eq ;<br><br>OR<br><br>lack of exercise;<br>obesity/heart disease/high blood pressure/ type ii diabetes/eq; | Allow drugs or named drug and their effect on health for 2 marks. | (2) |
| (b) | Any two from:<br>• production of bile reduced/affected;<br>• less fat emulsified/fat not emulsified;<br>• lipase less active/lipase unable to digest as much (fat)/slower lipase activity; | Allow bile not produced<br><br>Allow description of emulsification | (2) |
| (c) | blurred/misty vision/blindness/less light passing through the lens / less light reaching the retina; | | (1) |
| (d) | cerebrum/cerebellum/ cerebral cortex/cerebral hemispheres/hypothalamus/medulla (oblongata); | Allow any part of the brain that is affected by alcohol | (1) |
| (e) | Any three from:<br>• reduces release of neurotransmitters;<br>• reduced diffusion across synapse;<br>• less neurotransmitters bind to receptors on post-synaptic membrane;<br>• slows/ fewer transmission of impulses;<br>• slower reaction /reflexes; | | (3) |
| (f) | Any two from<br>• increased heart rate;<br>• increased breathing rate;<br>• pupil dilation;<br>• increase in blood sugar/glucose levels/ increased breakdown of glycogen to glucose;<br>• increase in blood flow to brain/muscles;<br>• increase in (aerobic) respiration (in muscles); | | (2) |

**Total for Question 3 = 11 marks**

## Type MS · Explain eye defects and how they are corrected or treated

### Answer 11 of 11 · Question 3 part a.iv · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part a.iv in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (a) | (i) | Part W - pupil;<br>Part X – retina; | 1 mark for each correctly named part | 2 |
| | (ii) | • sends (electrical) impulses from photoreceptors/rods/cones;<br>• to the brain/CNS;<br>OR<br>• sends (electrical) impulses from brain/CNS;<br>• to iris/ciliary muscles; | | 2 |
| | (iii) | lens becomes thinner/flatter/less convex;<br>less refraction; | allow less light refracted | 2 |
| | (iv) | Any two from:<br>• shape of cornea restored;<br>• light refracted, onto the retina/correctly/notmally;<br>• correct information sent to the brain;<br>• image/vision not blurred/clear; | | 2 |
| (b) | | • measure the size/diameter of the pupil;<br>• in a dark room;<br>• shine a light into the eye;<br>• measure/compare size/diameter of the pupil; | accept ref to different light intensities | 4 |

**Total Marks 12**

## Type MS · Explain eye defects and how they are corrected or treated

### Answer 11 of 11 · Question 3 part a.iv · 2 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part a.iv in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |